

PICOFLOW · DESIGN DECISION RECORD

PicoFlow recommendations accepted

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DOCUMENT CONTROL

Document ID	PCF-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/picoflow		
Source	docs/decisions/0002-recommendations-accepted.md @ a33f9df		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 acceptance of the TRL 3 recommendations (N1 to N5)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items N1 to N5); item O1 and the new item N6 remain proposed

■ Context

The TRL 3 session (docs/REVIEW.md , session "TRL 3", and PCF-DDR-001) left five new items marked "Proposed, awaiting Amish", each with a recommendation, and one older item (O1) with no recommendation. On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore decided as recommended. Items without one stay open. TRL 4 remains on hold by Amish's instruction, so any part of a decision that needs building, testing or purchasing is recorded but not carried out.

■ Options considered

Table 1. Items with a recommendation.

#	Item	Options	Recommendation
N1	Design-point penstock	110 mm; 125 mm	125 mm for a 20 m run
N2	R4, output at 1.0 m and 7 L/s	Keep 30 W and report not met; relax to 20 W	Relax to 20 W
N3	R2, flow range	Keep 5 to 15 L/s at every head; state 5 to 10 L/s at 1.0 m and 5 to 15 L/s from 2.0 m	State the range per head
N4	R14, turbine unit definition	Items 1 to 6 and 11; include the manifold and valve	Items 1 to 6 and 11
N5	R15, \$2 cost margin	Accept; trim cost (for example a printed bearing housing)	Accept, and firm prices with quotes before any build

■ Decision

- **N1.** Decided by Amish, 2026-09-25: go with recommendation. The design-point penstock is 125 mm drainage PVC (SN8, 117.6 mm bore) on a 20 m run. Pipe and branch loss falls from 23.6 % to 17.0 % of head; output into the battery rises from 77.2 W to 82.8 W and water to wire from 39.5 % to 43.2 %. The design-point inserts change from 34 mm to 33 mm. The penstock stays outside the kit budget; 20 m costs about \$140 instead of \$100.
- **N2.** Decided by Amish, 2026-09-25: go with recommendation. R4 is now 20 W or more at 1.0 m and 7 L/s (was 30 W). The calculation gives 26.3 W (was 23.9 W with 110 mm), so R4 is met.
- **N3.** Decided by Amish, 2026-09-25: go with recommendation. R2 is now 5 to 10 L/s at 1.0 m and 5 to 15 L/s from 2.0 m. The inserts pass 1.3 to 11.3 L/s at 1.0 m and 1.9 to 16.1 L/s at 2.0 m, so R2 is met.
- **N4.** Decided by Amish, 2026-09-25: go with recommendation. The turbine unit in R14 is items 1 to 6 and 11 (22.2 kg); the manifold and valve (5.2 kg) are pipework carried separately. R14 is met.

- **N5.** Decided by Amish, 2026-09-25: go with recommendation. The \$2 margin under the \$450 budget is accepted and R15 is recorded as met. Firming prices with supplier quotes is purchasing work, which is TRL 4 and on hold; it is recorded as decided but not done. `budget_usd` stays \$450.

Items that remain open (no recommendation, so they stay "Proposed, awaiting Amish"):

- **O1.** First site type, region and co-design partner, including confirmation that target sites carry about 5 to 15 L/s. Proposed, awaiting Amish.

New item raised while applying N1:

- **N6.** Inlet valve bore. `bom/bom.csv` lists a 90 mm (3 in) gate valve, while the calculation and the model treat the valve as full bore on the penstock. On a 125 mm line a 90 mm valve needs a reducer and an expander, which add loss and could erode the 2.8 W margin on R3. Options: (a) a full-bore gate valve matched to the penstock, re-priced in the kit; (b) keep the 90 mm valve and add the reducer losses to PCF-CAL-001. Recommendation: (b) first, to see whether R3 still holds, then (a) only if it does not. Proposed, awaiting Amish.

■ Consequences

- PCF-REQ-001 v0.4: R2 and R4 restated; R3, R4, R5 and R2 now met; R14 definition confirmed; R15 met with the margin accepted.
- PCF-CAL-001 v0.2 and `docs/04-calcs/sizing.py` : design penstock 125 mm (SN8, 3.7 mm wall); all results rerun. 13 met, 1 at risk (R12), 3 not verifiable at TRL 3, none not met.
- PCF-PRC-001 v0.4 and PCF-PRB-001 v0.4: numbers updated from PCF-CAL-001 v0.2.
- `cad/src/model.py` : penstock_od 110 to 125 mm, `jet_d` 32 to 33 mm; STEP and STL re-exported. PCF-DWG-001 moved from Rev P1 to Rev P2. Concept media regenerated.
- `bom/bom.csv` and `bom/bom-notes.md` : item 7 now uses a 125 x 90 mm reducing tee; item 9 is 125 mm pipe at about \$7.00 per meter (\$140.00, excluded). Kit total unchanged at \$448.00.
- `README.md` : concept numbers updated. `project.yaml` : no change needed (`budget_usd: 450` , `trl: 3` , `trl_target: 3`).
- PCF-DDR-001 v0.2: N1 to N5 marked as decided.